

AMUDEESHAN SRINIVASAN

AI Engineer · LLM Systems · Multi-Agent AI · RAG Pipelines

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PROFESSIONAL SUMMARY

AI Engineer specialising in local-first, privacy-preserving AI systems. Built ENMA a fully offline personal AI assistant with real-time voice, multi-model routing, hybrid RAG, tool orchestration, symbolic safety gates, and background automation entirely on local hardware using Python, Ollama, LanceDB, and Zig. Currently completing a Master of Artificial Intelligence at RMIT University (expected Jun 2026). Bringing 1.5 years of enterprise software engineering experience from Cognizant. Seeking an AI Engineer or LLM Systems role to build production-grade, trustworthy AI infrastructure.

TECHNICAL SKILLS

AI / LLM Engineering	LLMs, RAG, LangGraph, LangChain, ReAct, Transformers, Prompt Engineering, Vector Databases (LanceDB, pgvector), Agentic AI, Structured Outputs, Symbolic AI (Prolog, Clingo)
Voice & Perception	faster-whisper, Silero VAD, Piper TTS, RealtimeSTT, Wake Phrase Detection, Echo Suppression, V-JEPA (perception)
Backend & APIs	FastAPI, REST APIs, OpenAI-Compatible APIs, Tool Calling Systems, Async Pipelines, Zig
Machine Learning	PyTorch, TensorFlow, CNNs, RNNs, Embeddings, Cross-Encoder Reranking
Cloud & Infrastructure	AWS (EC2, SageMaker, EMR, Bedrock), Docker, Linux, Git, Ollama, SQLite, Jupyter
Programming	Python 3.12, Zig, JavaScript, Java, C#, SQL
Frontend	ReactJS, HTML, CSS

EDUCATION

Master of Artificial Intelligence

RMIT University | Melbourne, Australia

Mar 2024 – Jun 2026

Relevant coursework: LLMs, Deep Learning, NLP, Machine Learning, AI Systems Design

Bachelor of Engineering – Mechatronics

Anna University | Coimbatore, India

Aug 2018 – Jun 2022

AI PROJECTS

ENMA – Enhanced Neural Multi-Agent Assistant — Python 3.12 · Ollama · LanceDB · SQLite · Piper TTS · faster-whisper · Prolog · Zig · Clingo

- ▶ Built ENMA, a fully offline modular AI assistant spanning 27+ subsystems (agents, voice, RAG, memory, symbolic reasoning, perception, observability, runtime, and tooling) with 300+ commits, published on Hugging Face.
- ▶ Developed a low-latency real-time voice pipeline using Faster-Whisper, Silero VAD, wake-word detection, echo suppression, barge-in handling, and Piper/macOS TTS, achieving sub-2-second responses on the fast interaction path.
- ▶ Designed a turn-state routing and orchestration architecture that classifies intent and dispatches requests across 8 specialised routes (chat, coding, web research, deep reasoning, tool execution, etc.), preventing stale conversational context leakage between turns.
- ▶ Integrated and orchestrated multiple local LLMs through Ollama (Phi-4 Mini, Qwen3 14B, Qwen3-Coder 30B, GLM-4 Flash) with YAML-driven configuration management, privacy-gated cloud fallbacks, and hybrid RAG using LanceDB, embeddings, and SQLite memory.

- Engineered a multi-layer safety and runtime system including ToolGateway validation, symbolic Prolog permission checks, approval gates, runtime isolation, structured observability, and trusted web-research pipelines with caching, verification, and stale-result rejection.

Multi-Provider LLM API Router — Zig · AWS Bedrock · OpenRouter · Ollama

- Built an OpenAI-compatible /v1/chat/completions backend supporting dynamic routing across Ollama, OpenRouter, and AWS Bedrock — enabling zero-code provider switching.
- Developed modular provider adapters with normalised request/response handling, structured error management, and environment-based configuration.
- Implemented a lightweight HTTP backend from scratch without external frameworks, keeping the binary footprint minimal for edge deployment.

Tool-RAG AI Assistant — Python · FastAPI · LangChain · Vector DBs

- Developed a local AI assistant combining ReAct reasoning, structured outputs, and API-backed tool execution for multi-step query resolution.
- Implemented a hybrid retrieval pipeline (embedding-based vector search + cross-encoder reranking) to improve answer relevance over naive top-k retrieval.
- Optimised local inference using quantised models and KV-cache tuning, reducing memory footprint while maintaining response quality.

Zig Evaluation & Benchmarking Framework — Zig · LLM Evaluation

- Built a lightweight evaluation and benchmarking framework for testing LLM and agent workflows with structured output validation.
- Developed benchmarking utilities for comparing orchestration patterns, response quality, and runtime behaviour across AI systems.

PROFESSIONAL EXPERIENCE

Programmer Analyst

Feb 2022 – Sep 2023

Cognizant Technology Solutions | Coimbatore, India

- Contributed to Gilead's GCAP SharePoint migration — a large-scale enterprise modernisation initiative — delivering frontend components in ReactJS, HTML, and CSS.
- Developed and maintained enterprise web applications using .NET, C#, and MS SharePoint, supporting cross-team agile delivery workflows.
- Resolved production issues and supported system integration activities, reducing downtime through systematic debugging and root-cause analysis.

PG — AI Systems / Agent Engineering (Industry Capstone Project)

March 2026 – June 2026

Scalabs | Remote, Germany

- Contributing to a lightweight AI coding-agent harness in Zig focused on ReAct, tool-use loops, plan-then-execute, and multi-agent orchestration patterns.
- Working on OpenAI-compatible LLM APIs, streaming response handling, context management, and agent workflow infrastructure.
- Collaborating in a remote EU engineering environment using GitHub workflows, async standups, Discord coordination, and open-source development practices.
- Researching low-overhead terminal-based AI agent architectures as alternatives to React/Ink-based coding agents.

OPEN SOURCE & ADDITIONAL WORK

- Open-sourced AI infrastructure and agentic AI systems on GitHub and Hugging Face, including multi-agent assistants, routing infrastructure, and evaluation tooling.
- Built OpenAI-compatible LLM routing systems supporting Ollama, AWS Bedrock, and OpenRouter integrations.
- Experimented with symbolic AI, vector-memory pipelines, structured outputs, and production-grade AI orchestration workflows.